



Introduction to Express JS

1. Express JS - need and features
2. Handling requests
3. Routing parameters

- Express.js is a web application framework which provides various features that make web application development fast and easy.
- Features of Express JS
 - Faster server side development
 - Routing
 - Middleware
 - Serving static content
 - Templating
 - Building REST server
 - Communicate with databases

- Advantages
 - Single language for backend and frontend development.
 - Faster with databases like MySQL and MongoDB
 - Dynamic rendering of HTML is possible by template engines
- Limitations
 - Express JS is totally based on use of callbacks
 - Error messages are not user friendly
- Installation
 - Express.js can be installed using npm utility locally or globally (using -g)

- Once request is received from client, server(node application) will choose the corresponding route handler
- A route handler is a function that will be executed every time the server receives a particular request, defined by HTTP method.
- Syntax:

```
app.method('request url', callback){  
  ...  
}
```
- Example

```
app.get('/', (req, res) => {  
  res.send('root')  
})
```

- Route path
 - Route path along with http method is used by the client to make a request
 - Route paths can be strings, string patterns, or regular expressions.
- General route
 - A route handler which can handle any request url
 - Should be written at the end of all routes when none of the route matches, this general route gets executed

- Users can add a specific variable(s) to the request URL.
- We can use : symbol in request url for specifying route parameter in request url in the route definition.
- Route handler function can use req.params object for reading(capturing) the request parameter.